Constraining Magma Reservoir Characteristics Beneath the Summit of Olympus Mons on Mars

Henry Peterson

October 19, 2022

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Abbreviations

CTX Context Camera 3

LIP Large Igneous Province 2

MOLA Mars Orbiter Laser Altimeter 1

MRO Mars Reconnaissance Orbiter 3

OM Olympus Mons 1–3

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1 Introduction

1.1 Motivation

The study of Olympus Mons (OM) contributes to a more comprehensive understanding of magma reservoirs than could be achieved with a solely Earth-based inquiry. Compared to Earth, the unique challenges of studying Martian volcanoes are obvious: the inability to make in-situ measurements and the lack of evidence for ongoing eruptive activity stand out. However, there are important advantages as well. Most notably, surface deformation signatures are well preserved in the Martian environment over hundreds of millions of years, especially at high elevations where weathering and erosion are far less active. This allows for a more comprehensive picture of edifice deformation than is commonly available on Earth, where such edifices are rare and often poorly preserved. Additionally, the unique Martian conditions that yield the largest volcanoes in the solar system broaden our perspective of how volcanism works in general. Finally, previous work investigating Olympus Mons and nearby Tharsis Montes has already yielded clear evidence for surface deformation. Specifically, numerous lava flows (which generally move straight downhill during eruptions) are misaligned with the local topographic downhill direction measured by the Mars Orbiter Laser Altimeter (MOLA). This finding suggests subsequent deformation, most likely because of reservoir activity at summit of Olympus Mons. The current project's role is to provide a quantitative link between observed surface deformation and inferred reservoir activity.

1.2 Research Objectives

The following question will guide this research project: what combination of magma reservoir characteristics (e.g., geometry, depth) and magma injection event parameters best accounts for both

1. observed lava flow orientations
2. modern summit topography

My hypothesis is that significant pressurization of a shallow, relatively small-footprint magma reservoir centered near the modern summit will best explain the observed discrepancy between the observed surface and the older deflected lava flows. To test this hypothesis, I will

1. map long lava flows near the surface
2. constrain their deflection from the modern topographic surface, i.e. the extent to which they are not flowing 'downhill'
3. use crater dating to estimate relative flow ages to constrain a timeline of deformation
4. run numerical models of the volcano and summit reservoir to assess the conditions needed to generate observed topographic deflections.

1.3 Literature Review

1.3.1 Geologic Overview of OM

The volcanic history of Mars differs from that of the Earth in many ways. On Earth, billions of years of tectonic activity give rise to distinctive volcanic provinces including mid-ocean ridges, island & continental arcs, hotspot chains, and Large Igneous Provinces (LIPs). Ridges and arcs have no clear analogues on Mars; hotspots and LIPs have analogous expressions but reflect differing thermal, mechanical, and chemical properties.

Large-scale differences that emerge in the comparitive study of Earth and Mars are reflected in the respective geologic timescales used for each planet.

1.3.2 The Lava Flow Alignment Method

1.4 Guiding Assumptions

2 Methods

2.1 Selection and Preparation of Published Data

The Context Camera (CTX) aboard the Mars Reconnaissance Orbiter (MRO) satellite has captured visible greyscale swaths at roughly 6 m resolution over the vast majority of the Martian Surface. These have been stiched together into a mosaic covering the entire Martian surface Dickson et al. 2018. This product (hereafter, "CTX mosaic") will be used to visually identify and map lava flows, flow margins, contacts, and flow channels.

2.2 Definition of Scope

2.2.1 Lava Flow Mapping

The global CTX mosaic was used to identify lava flows near the summit of OM. Where possible, lobate flows were outlined and mapped as polygons in ArcGIS Pro. From these polygonal features, flow centerlines were created. Channels were mapped as lines where flow margins were not visible. The "Polygon to Centerline" tool was used to derive linear flow paths from the polygonal flow outlines.

2.2.2 Point Grid

The "Generate Tesselation" tool was used to construct a grid of squares (polygon features) with area (FIX) $0.36\,\mathrm{km}^2$ over the extent of the lava flows. Of those squares whose centers lie within a flow polygon, a point feature was created using the "Polygon to Point" tool.

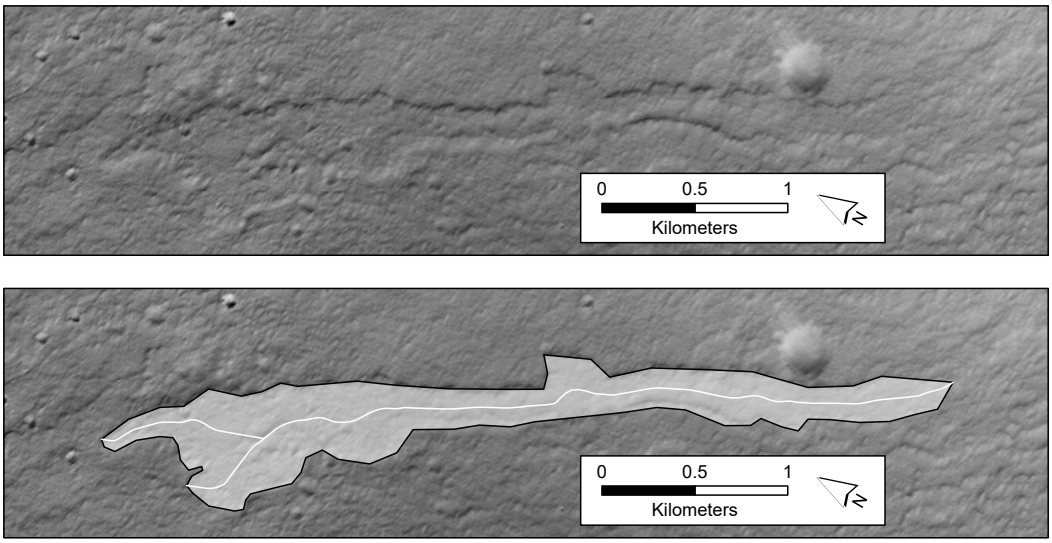

Figure 2.1: A lobate flow is first identified by relatively clear boundaries at each margin (above). The flow is then mapped as a polygon (outlined in black) and centerline (white).

The overall effect of this procedure is to generate a square grid of points that spans the entire area covered by the mapped lava flows. To each of these points will be assigned several varaibles representing various descriptive parameters of the modern and paleo surface.

2.3 Surface Parameters

I run the "Surface Parameters" tool on the 200 m DEM surface twice, once to produce a slope raster (in degrees) and once to produce an aspect raster (also in degrees, measured clockwise from due north). In both cases, I used a quadratic local surface type with a neighborhood radius of 2 km. I then used the "Add Surface Informaton" tool to assign these values to each feature in the point grid.

2.4 Paleosurface Parameters

2.4.1 Paleoaspect

To assign paleoaspect, I first found the angle from each point to the nearest flow centerline or channel. This angle is perpendicular "Near" tool was used to assign the angle from the nearest flow path to each grid point. This angle is perpendicular to the paleoaspect (downhill direction) recorded by the flow. Thus, 90° is added to the "Near Angle", and the result is stored modulo 360° to avoid counting extra rotations in the final aspect direction. Since points in the grid appear on both sides of the centerline, the stored angle may be parallel or anti-parallel to the actual direction of flow. Based on visual inspection of flow morphologies, I assumed that all flows were oriented such that the paleosurface sloped downhill away from the caldera, not toward it. Thus, I used the "Near" tool again to determine the angle from the caldera to each grid point. If the difference between this angle and the stored aspect angle (absolute value) exceeded 180°, I recalculated the aspect value by adding 180 (again modulo 360°).

2.4.2 Paleoslope

It is not possible to assign a paleoslope directly from the lava flows like we can with the paleoaspect. Instead, I chose to place a constraint on the paleoslope by assuming the minimum possible surface attitude change between flow emplacement and the modern surface. Equivalently, since each surface attitude is equivalent to a unique unit normal vector, I sought to minimize the angular distance between the normal vectors corresponding to the

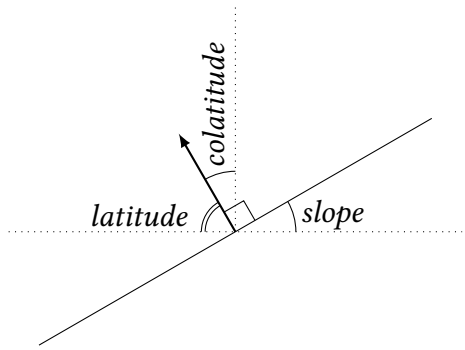

Figure 2.2: ArcGIS Pro calculates the slope of a tilted Digital Elevation Model surface, which is equal to the colatitude of the normal vector (complement of latitude). This angle is also known as "dip."

modern and paleosurfaces. From the previous steps, we have a grid of point features, each with attributes including "SLOPE" (S), "ASPECT" (A), and "PALEO_ASPECT" (A′). Note that these values are stored in degrees; corresponding lowercase letters are equivalent values converted into radians for trigonometric calculations.

The basic strategy for solving this optimization problem is as follows. First, the slope and aspect of the modern surface are plotted on the surface of a unit sphere as a colatitude and longitude, respectively. This point on the sphere corresponds to the unit vector normal to the surface. Note that the sine of a colatitude is equal to the cosine of a latitude, and vice versa; this substitution will be used in the following equations.

For the paleosurface, only the aspect is known. Instead of a unique point on the sphere, we have a line of longitude. To minimize the angular distance between the two surfaces, we are thus looking for the shortest great circle arc between the modern surface point and the paleosurface line.

To achieve this optimization and find the nearest intersection point, the sphere surface is converted to a gnomonic projection: one in which all great circles are mapped to straight lines in the cartesian plane. Then, we find the line perpendicular to the paleoaspect line which contains the modern surface point. The radial distance of this intersection point from the origin is reprojected onto a spherical colatitude, which is equivalent to the surface slope.

The equations take the following form. First, the line[1] corresponding to the paleoaspect direction in polar coordinates:

[1]This line is really a ray from the origin, because any negative values of r for a given θ in polar coordinates will correspond to a negative slope, which would have been captured by a positive slope for the opposite aspect. This case will be handled later.

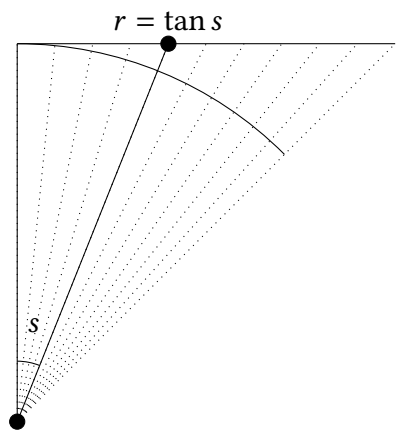

Figure 2.3: The gnomonic projection is used to transform aspect (longtitude of normal vector) and slope (colatitude of normal vector) into cartesian coordinates x and y such that great-circles are preserved as straight lines. A cross-section along a single line of longtitude is displayed here. In the transformation to polar coordinates r and θ, longitude is preserved ($a \longrightarrow \theta$) while colatitude is distorted ($\tan s \longrightarrow r$)

$$\theta = a'. \tag{2.1}$$

In cartesian coordinates:

$$y = x \cot a'. \tag{2.2}$$

The point corresponding to the modern surface in polar coordinates:

$$[r, \theta] = [\tan s, a]. \tag{2.3}$$

In cartesian coordinates:

$$[x, y] = [\tan s \sin a, \tan s \cos a]. \tag{2.4}$$

Recall that we are looking for the shortest path between the longitudinal line on the sphere (reflecting paleoaspect with unknown paleoslope) and the point (reflecting known moderns slope and aspect). The gnomonic projection was used because great-circle (shortest distance) paths on a sphere are projected as straight lines. Therefore, we need to find the shortest path between the cartesian line and the cartesian point. The point where this path intersects the line will then place a constraint on paleoslope; i.e., it answers the question "which paleoslope (s') yields the points $[s', a']$ given a known a' which is

closest to $[s, a]$." The closest path from the point to the line is perpendicular to the line. Perpendicular to slope $m = \cot a'$ is slope $m = -\tan a'$. The full point-slope form of the line is:

$$y - \tan s \cos a = -\tan a'(x - \tan s \sin a). \tag{2.5}$$

We are looking for the intersection point of these two lines, the point that satisfies both 2.2 and 2.5. Substituting from 2.2 to find the x-coordinate of the intersection point:

$$x \cot a' - \tan s \cos a = -\tan a'(x - \tan s \sin a)$$
$$x = \frac{\tan s(\cos a + \tan a' \sin a)}{\cot a' + \tan a'} \tag{2.6}$$

Since the math module in python does not include cotangent, I used the following exact formula in ArcGIS, which is equivalent to 2.6:

$$x = \sin(2a') \tan s(\cos a + \tan a' \sin a)/2 \tag{2.7}$$

And from 2.2, the $y-$coordinate of the intersection point:

$$y = x \cot a' \tag{2.8}$$

The polar coordinate r is given by $\sqrt{x^2 + y^2}$; which is reprojected back onto the sphere, giving the paleoslope:

$$s' = \arctan \sqrt{x^2 + y^2} \tag{2.9}$$

The final result of this sequence of calculations is to identify the paleoslope which, having been deformed to give the modern surface, would have required the least possible angular change in attitude.

2.4.3 Deflection

Before proceeding, we should check that the deflection (defined $A - A'$) does not have a magnitude greater than 90°. If it does, then this calculation has produced a nearest uphill slope, rather than downhill slope, which is clearly not the desired behavior. Therefore, at this stage any grid points with such deflections have their paleoslopes redefined to zero. Alternatively, a minimum slope required for lava flow can also be set here.

2.4.4 Deformation

From 2.4.2 we have generated two points on a spherical surface, each corresponding to a unit vector normal to a unique surface. The angular distance between these vectors (also,

the angle between the two planes) defines the attribute "DEFORM" for each grid point. The formula[2] is:

$$\arccos\left[\cos s \cos s' + \sin s \sin s' \cos(a - a')\right] \tag{2.10}$$

converted into degrees for comparison with other attributes.

2.4.5 Elevation Change

2.4.6 Volumetric Inflation

[2]The trigonometric functions cos and sin are swapped for s and s' because these are colatitudes, not latitudes as the most common formula assumes.

References

Dickson, J. L., Kerber, L., Fassett, C. I., & Ehlmann, B. L. (2018). A global, blended CTX mosaic of Mars with vectorized seam mapping: A new mosaicking pipeline using principles of non-destructive image editing.